Tribhuvan University
Faculty of Humanities and Social Science

A PROPOSOL REPORT

ON

RESUME ANALYZER & JOB MATCHING SYSTEM

Submitted to

Department of Computer Application

Pascal National College

In partial fulfillment of the requirements for the degree of Bachelors in Computer Application

Submitted by

Sangam Subedi | 6-2-1226-21-2021

April 2026

Table of Contents

1. Introduction

In today's competitive job market, having a well-structured and optimized resume is very important for job seekers to get shortlisted by employers. Many candidates, especially students and fresh graduates, struggle to create effective resumes that meet industry standards and pass through Applicant Tracking Systems (ATS). Due to lack of proper guidance, resumes often miss important keywords, have poor formatting, or fail to highlight relevant skills and experiences, which reduces their chances of getting selected.

This project, titled "Resume Analyzer and Job Matching System", is developed to address these problems by providing an intelligent platform that helps users evaluate and improve their resumes. The system allows users to upload their resumes and compares them with job descriptions to determine how well they match. It analyzes different aspects such as keywords, skills, content quality, and overall structure, and provides a score along with suggestions for improvement.

Unlike many modern systems that rely heavily on advanced AI generation tools, this project focuses on implementing fundamental algorithms and techniques such as text preprocessing, keyword extraction using TF-IDF, cosine similarity for job matching, and rule-based recommendation systems. This approach helps in understanding the core logic behind resume analysis while keeping the system simple, cost-effective, and efficient.

The main objective of this project is to build a user-friendly web-based application that not only evaluates resumes but also guides users in improving them based on data-driven insights. This system will be especially useful for students and job seekers who want to enhance their resumes and increase their chances of securing employment opportunities.

2. Problem Statement

In today's competitive job market, many students and job seekers struggle to create effective resumes that meet industry standards and pass Applicant Tracking Systems (ATS). Due to lack of proper knowledge, resumes often miss important keywords, have poor structure, or fail to highlight relevant skills and experiences, reducing the chances of getting shortlisted. Additionally, many applicants apply for jobs without knowing how well their resumes match the job requirements.

There is a lack of simple, cost-free, and user-friendly tools that can analyze resumes and provide clear, understandable feedback. Existing solutions are often complex or paid, making them less accessible. Therefore, this project aims to develop a Resume Analyzer and Job Matching System that evaluates resumes, identifies gaps, and provides meaningful suggestions using basic algorithms, helping users improve their resumes and increase their chances of selection.

3. Objectives

- To extract and process resume content from PDF and DOCX files.
- To identify important keywords using algorithms like TF-IDF.
- To identify missing skills and keyword gaps in the resume.
- To calculate a resume score based on different factors like keywords, skills, and content quality.

4. Background Study and Literature Review

Background Study

In recent years, the job market has become highly competitive, and recruiters increasingly rely on digital systems such as Applicant Tracking Systems (ATS) to filter and shortlist candidates. Because of this, the structure, formatting, and keyword optimization of a resume play a very important role in determining whether a candidate gets selected or rejected. Many students and job seekers, especially fresh graduates, struggle to create effective resumes due to a lack of proper guidance and understanding of industry requirements.

Existing resume-building platforms and online tools provide basic templates and formatting assistance, but most of them do not offer detailed analysis or personalized feedback on resume quality. Some advanced platforms use artificial intelligence for resume evaluation, but they are often complex, expensive, or not transparent in their evaluation methods. This creates a gap for a simple, cost-free, and understandable system that can help users analyze their resumes and improve them based on clear insights.

This project, the Resume Analyzer and Job Matching System, is developed by studying these existing limitations and applying fundamental concepts of data processing and information retrieval. Techniques such as text preprocessing, keyword extraction using TF-IDF, and similarity measurement using cosine similarity are commonly used in document analysis and natural language processing tasks. By implementing these techniques in a simple web-based system, this project aims to provide meaningful resume evaluation and job matching functionality in an accessible and user-friendly manner.

Literature Review

The authors in the research paper on the topic “A Smart Resume Analyser and Recommendation System” stated that Every year millions of students are introduced to the actively competitive environment of job seekers. Finding a job that not only pays well but also meets your requirements is a tedious process, which requires countless hours of research from numerous resources across the Internet. There are many sources that are to be considered simultaneously, most of which are outdated or biased. In this situation knowing which sources to trust and getting an unbiased yet knowledgeable understanding of the Jobs is of utmost importance. With the growth in the Job market there’s also been limitless expansion in the data which connects several technical and non-technical, personality based, experience based, skill set based, attitude-based parameters to each other to create thousands of large high complexity datasets which are impossible for the common person to analyze and benefit from. This is where machine learning algorithms and data visualization techniques play a significant role in not only analyzing thousands of datasets but also filtering out the outliers and identifying a trend which it then depicts in simplified forms for the user to understand [1].

The authors in a research paper on the topic “Smart Resume Analyzer” mentioned that The Resume Analyzer system is a method of applying text mining to analyze resumes received by the company by utilizing keyword matching algorithms. It is being implemented and evaluated. We have come to believe that technology advancements are reaching unprecedented heights these days. As a result, many people now have a lot of good opportunities for employment. However, each business operates in a unique manner. They need someone with a certain skill set for this unique way of functioning. These hiring decisions are made exclusively on the basis of the skill set that the candidate has indicated on their resume. We can now see that hundreds of individuals are watching a certain activity. Going through these people’s resumes by hand takes a lot of time and is not very environmentally friendly because there is a chance that mistakes will be made during human intervention. As a result, we have suggested a project to create all of the resumes ahead of them and in accordance with commercial enterprise requirements. We will build a resume scanner and analyzer using machine learning in this project. These days, the majority of agencies filter resumes containing necessary keywords using an ATS (Automatic Tracking System). However, there isn't a tool like that on the student side to help him strengthen his resume. As a result, we are developing software that will accept candidate or student resumes and create a file mostly based on them [2].

Additionally, web-based resume building platforms such as Resume.io provide users with automated resume creation tools and templates, while focusing mainly on formatting and design rather than deep analytical evaluation [3]. Similarly, Resume.com offers free resume-building services with basic guidance and templates, but lacks advanced features such as job matching and skill gap analysis [4]. These limitations highlight the need for a more transparent and analytical system that focuses on resume evaluation and job matching using algorithmic approaches.

5. Methodology

This project will be developed using the Agile methodology, which follows an iterative and incremental approach where the system is built in small parts called sprints. In the beginning, requirement analysis is conducted to identify all functional and non-functional requirements such as resume upload, text extraction, keyword analysis, job matching, scoring, and suggestion generation. These requirements are then divided into smaller tasks and prioritized for development. In each sprint, design, implementation, and testing of selected features are carried out simultaneously, for example implementing resume upload and parsing in one sprint and keyword extraction and job matching in another using algorithms like TF-IDF and cosine similarity. Continuous feedback and testing are performed throughout the development process to improve the system step by step. Finally, after completing all sprints, the modules are integrated, tested as a complete system, and delivered as a fully functional Resume Analyzer and Job Matching System.

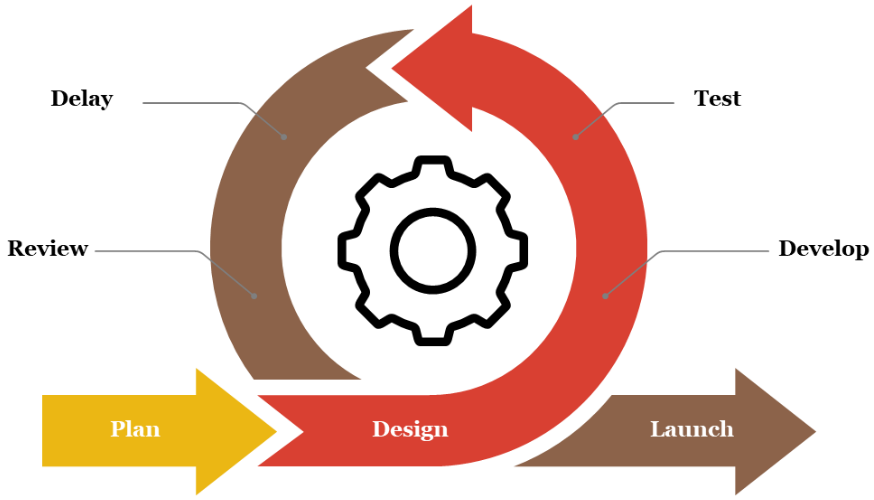

Figure 1: Agile Methodology

System Analysis

a. Requirement Identification

i. Functional Requirements:

- The system shall allow users to register and log in securely.
- The system shall extract text from uploaded resumes.
- The system shall allow users to input job descriptions.
- The system shall evaluate resume quality based on predefined criteria.
- The system shall identify missing skills and keyword gaps.

ii. Non-Functional Requirements:

- The system should process and analyze resumes quickly.
- The system should be easy to use for non-technical users.
- The system should provide accurate and consistent results.
- User data and resumes should be protected and not publicly accessible.
- The system should work on different browsers and devices.

b. Feasibility Study

i. Technical Feasibility:

The project is technically feasible as it uses widely available technologies such as React for frontend, Express for backend and PostgreSQL (Supabase) for database.

ii. Operational Feasibility:

The system is easy to operate and designed with a simple user interface. Users can upload resumes and receive analysis results without requiring technical knowledge. This makes the system suitable for students and job seekers.

iii. **Economic Feasibility:**
The project is cost-effective as it uses free and open-source tools and platforms. Services like Firebase and Supabase offer free tiers, and no expensive hardware or software is required, making it suitable for a student-level project.

Working Mechanism of the System:

The Resume Analyzer and Job Matching System works through a sequence of steps that process user input, analyze resume content, and generate meaningful results. Initially, the user logs into the system and uploads their resume in PDF or DOCX format. The system then extracts the textual content from the uploaded file using parsing techniques. After extraction, the text undergoes preprocessing, which includes converting text to lowercase, removing stopwords, and tokenizing the words to make the data suitable for analysis.

Once the text is cleaned, the system applies the TF-IDF algorithm to identify important keywords from both the resume and the job description provided by the user. These keywords are then converted into numerical vectors. After vectorization, the cosine similarity algorithm is used to compare the resume with the job description and calculate a similarity score, which represents how well the resume matches the job requirements.

In addition to job matching, the system evaluates the overall quality of the resume using a scoring mechanism based on factors such as keywords, skills, and content structure. It also performs skill gap analysis to identify missing skills by comparing the resume with the job requirements. Based on this analysis, the system generates rule-based suggestions to help users improve their resumes. Finally, all results, including match score, resume score, missing skills, and suggestions, are displayed to the user through an interactive interface, making it easier for them to understand and enhance their resume.

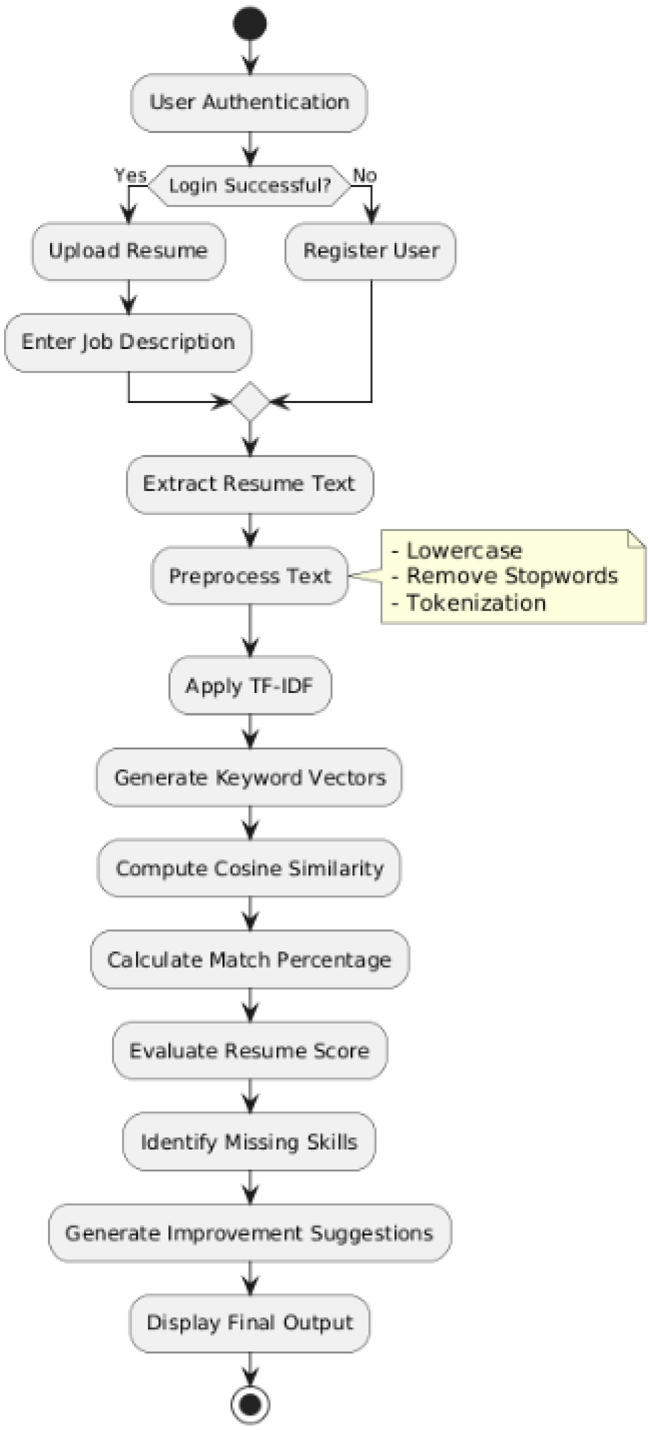

Figure 3: Working Mechanism

Data Flow Diagram (DFD):

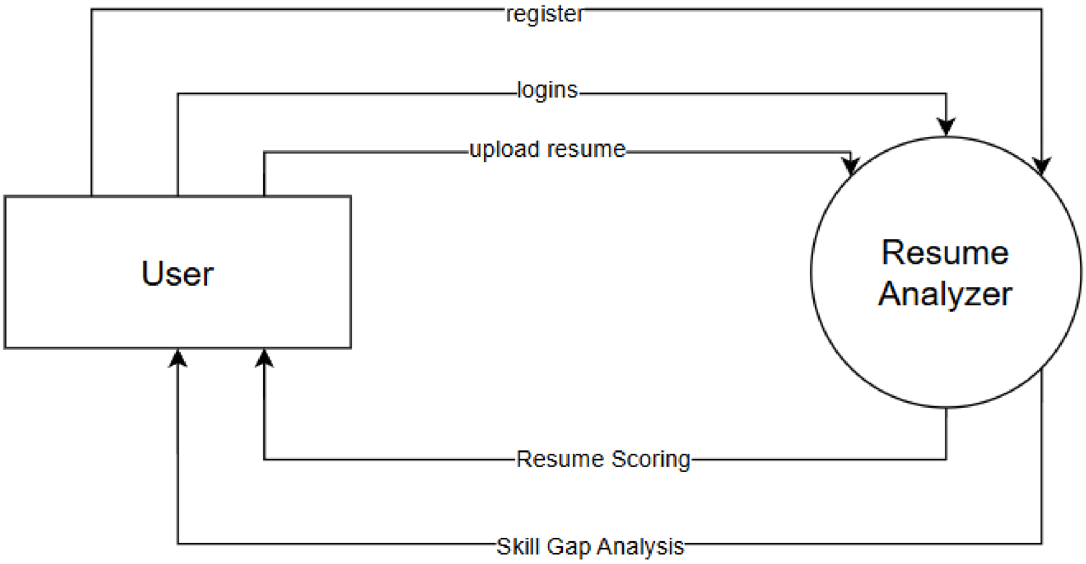

Figure 4: DFD

Algorithm Details

TF-IDF Algorithm:

TF-IDF is a statistical technique used to identify the importance of words in a document relative to a collection of documents. In this project, it is used to extract meaningful keywords from both the resume and the job description. The algorithm works by calculating two values: Term Frequency (TF), which measures how often a word appears in a document, and Inverse Document Frequency (IDF), which measures how unique or rare the word is across all documents. Words that appear frequently in a specific document but not in many other documents receive higher TF-IDF scores, making them more important for analysis. By applying TF-IDF, the system can focus on relevant skills and terms such as programming languages or technologies while ignoring common words, which helps improve the accuracy of resume evaluation and job matching.

$$\text{TF} = \frac{\text{Number of times words appear}}{\text{Total words in document}} \text{ and IDF} = \log\left(\frac{\text{Total Documents}}{\text{Documents Containing Words}}\right)$$

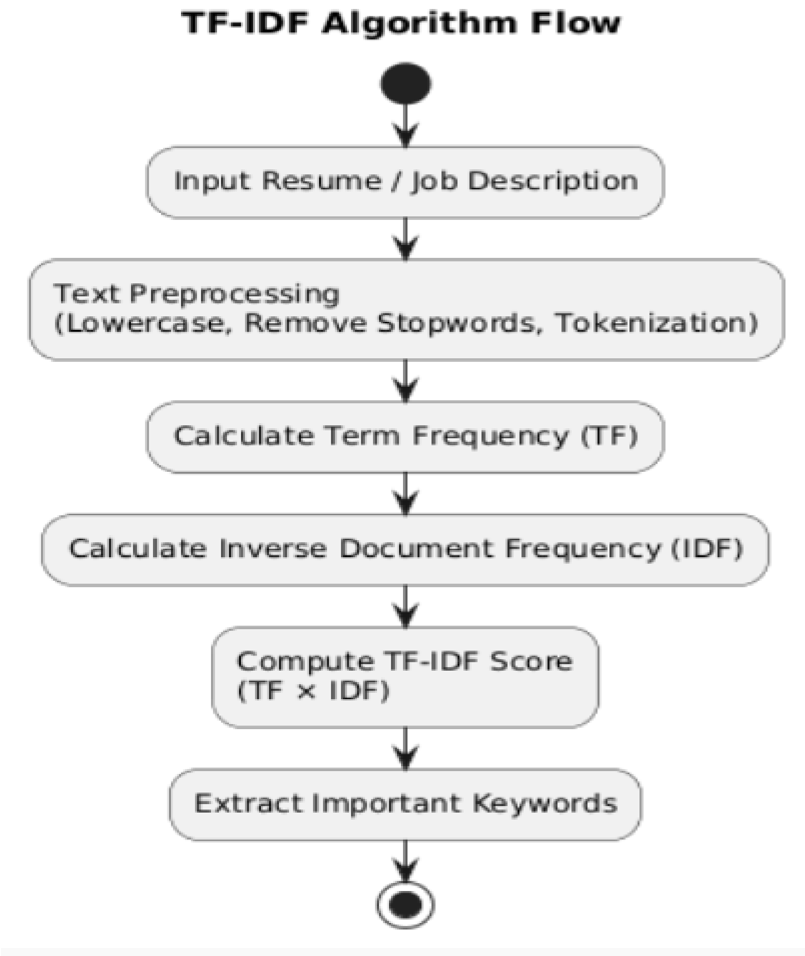

Figure 5: TF-IDF Working Diagram

Cosine Similarity Algorithm:

Cosine Similarity is a mathematical algorithm used to measure the similarity between two documents by comparing them as vectors in a multi-dimensional space. In this project, it is used to determine how closely a resume matches a given job description. After converting both the resume and job description into numerical vectors (using TF-IDF), the cosine similarity algorithm calculates the cosine of the angle between these two vectors. The result ranges from 0 to 1, where 1 indicates a perfect match and 0 indicates no similarity. A higher similarity score means that the resume contains more relevant keywords and skills required for the job. This algorithm helps in ranking candidates and providing a clear match percentage, making it easier for users to understand their suitability for a specific job role.

$$\cos(\theta)=\frac{A\,.\,B}{|A|\,.\,|B|}$$

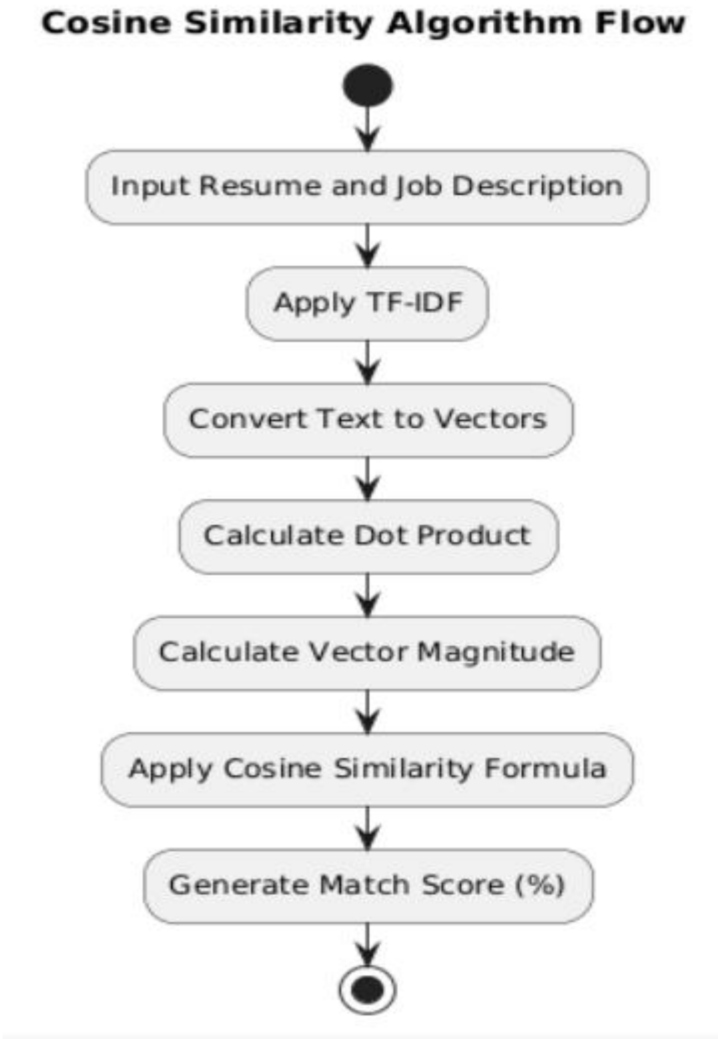

Figure 6: Cosine Similarity Flow Diagram

6. Gantt Chart

	Start Date	End Date	1	2	3	4	5	6	7	8	Status
Research and Planning	1-April	4-April	■								Ongoing
Requirement Gathering	5-April	8-April		■							Upcoming
System Design	9-April	20-April			■	■					Upcoming
System Development	21-April	10-May					■	■			Upcoming
Review	11-May	18-May							■		Upcoming
Testing	19-May	28-May								■	Upcoming

Figure 7: Gantt Chart

7. Expected Outcome

The expected outcome of this project is the development of a fully functional Resume Analyzer and Job Matching System that allows users to upload their resumes and receive detailed analysis results. The system will successfully extract and process resume content, identify important keywords, and compare the resume with a given job description to generate a match score. Users will be able to clearly understand how well their resume aligns with job requirements.

References

[1] R. S. T. S. a. S. B. B. Sameer Shaikh, "Job Analista: A Smart Resume Analyser and Recommendation System," *International Journal of Innovative Research in Science, Engineering and Technology,* July 2024.

[2] K. R. S. a. P. Kumar, "Smart Resume Analyzer: An Automated Approach for Recruitment Process," in *2025 International Conference on Pervasive Computational Technologies (ICPCT)*, 2025.

[3] Resume.io, "Online Resume Builder: Professionally Tested Templates," 2023. [Online]. Available: https://resume.io/. [Accessed 24 10 2023].

[4] Resume.com, "Resume.com: Build a Professional Resume for Free," 2025. [Online]. Available: https://www.resume.com/.